# USB Insider Forensics: Detecting Data Exfiltration via Removable Media on Windows

**Case:** INSIDER-USB-001
**Investigator:** Student Investigator

## Executive Summary

This controlled investigation evaluated device, filesystem, Registry/event, and user-activity o
bservations. H3 is **SUPPORTED** (HIGH). The assessment relies on corroboration rather than a s
ingle artifact.

## Investigation and Research Questions

Did employee01 copy confidential company data to an external USB device, when, which files, and
 what independent evidence supports the conclusion? Can Windows artifacts distinguish legitimat
e USB use from simulated insider leakage?

## Scope and Evidence Sources

Offline analysis of Windows 10/11 artifact exports: a controlled two-scenario dataset using fic
tional Example Corporation data, optionally complemented by read-only live triage of a real PC.

## Evidence Inventory

- EVD-EXPORT-001 | real-run (artifact export bundle) | SHA-256 `20f66c1aac9a4455333462f63d0baa1
7875c2c1269673e7337695766c978ef23` | 9300 bytes
- Imported artifact export bundle; original source files preserved and hashed for analysis prov
enance.

Raw exports are preserved separately; normalized records retain their original payload and prov
enance identifiers.

## Acquisition Methodology and Hash Verification

The included demo hashes the synthetic export bundle for provenance. Real evidence is either co
llected read-only with the triage collector, which records a SHA-256 manifest, or acquired as a
 powered-off disk and USB image that is hashed, logged in custody and examined only as a verifi
ed copy.

## Artifact, USB, User, and Filesystem Analysis

- META-EXP-A: simulation_metadata / None / n/a (LOW)

- A-0001: user_session / logged_on / n/a (MEDIUM)

- A-0002: usb / connected / Seagate Expansion SCSI Disk Device (MEDIUM)

- A-0003: filesystem / created / Readme.txt (MEDIUM)

- A-0004: lnk / accessed / Readme.txt (MEDIUM)

- META-EXP-B: simulation_metadata / None / n/a (LOW)

- B-0001: user_session / logged_on / n/a (MEDIUM)

- B-0002: usb / connected / Seagate Expansion SCSI Disk Device (MEDIUM)

- B-0003: filesystem / created / Employee_Salaries.xlsx (MEDIUM)

- B-0004: filesystem / created / Customer_Database.csv (MEDIUM)

- B-0005: archive / archived / Project_Files.zip (MEDIUM)

- B-0006: lnk / accessed / Employee_Salaries.xlsx (MEDIUM)

- B-0007: lnk / accessed / Customer_Database.csv (MEDIUM)

- B-0008: lnk / accessed / Project_Apollo.pdf (MEDIUM)

- B-0009: recycle_bin / deleted / Project_Apollo.pdf (MEDIUM)

## Unified Timeline

- 2026-10-21T20:47:29+05:30 — A-0001 — Interactive user logon

- 2026-10-21T20:47:29+05:30 — B-0001 — Interactive user logon

- 2026-10-21T21:02:07+05:30 — A-0002 — Last connection of USB storage device

- 2026-10-21T21:03:08+05:30 — A-0003 — Destination-side file on removable media

- 2026-10-21T21:03:08+05:30 — A-0004 — Shortcut created when the file was opened

- 2026-10-21T21:32:36+05:30 — B-0002 — Last connection of USB storage device

- 2026-10-21T21:32:57+05:30 — B-0003 — Destination-side file on removable media

- 2026-10-21T21:32:57+05:30 — B-0004 — Destination-side file on removable media

- 2026-10-21T21:32:57+05:30 — B-0006 — Shortcut created when the file was opened

- 2026-10-21T21:32:57+05:30 — B-0007 — Shortcut created when the file was opened

- 2026-10-21T21:32:57+05:30 — B-0008 — Shortcut created when the file was opened

- 2026-10-21T21:32:58+05:30 — B-0005 — Archive present on removable media

- 2026-10-21T21:32:59+05:30 — B-0009 — File moved to the Recycle Bin

- unknown — META-EXP-A —

- unknown — META-EXP-B —

## Correlations

- CORR-001 USER_SESSION (MEDIUM): User-session artifact observed

- CORR-002 USB_SESSION (MEDIUM): Removable-device connection artifact observed; No disconnect
rtifact; session end is unknown

- CORR-003 USER_SESSION (MEDIUM): User-session artifact observed

- CORR-004 USB_SESSION (MEDIUM): Removable-device connection artifact observed; No disconnect
rtifact; session end is unknown

- CORR-005 POSSIBLE_TRANSFER_SESSION (HIGH): USB device activity observed; Sensitive-file act
ity occurred within the configured time window; Employee user-session artifact corroborates tem
poral association; Destination-side file artifact was observed on removable media; Archive acti

vity occurred in the same experiment

- CORR-006 SENSITIVE_FILE_ACTIVITY (MEDIUM): Project-classified sensitive files have activity a rtifacts

- CORR-007 ARCHIVE_ACTIVITY (MEDIUM): Archive creation artifact observed; intent is not establi shed

- CORR-008 POSSIBLE_CONCEALMENT (LOW): Rename or deletion artifact observed; Artifact presen alone does not establish malicious intent

## Hypothesis Evaluation

### H1 — SUPPORTED (HIGH)
- USB connection artifacts support device presence

### H2 — SUPPORTED (MEDIUM)
- User-associated sensitive-file artifacts were observed

### H3 — SUPPORTED (HIGH)
- USB device activity observed
- Sensitive-file activity occurred within the configured time window
- Employee user-session artifact corroborates temporal association
- Destination-side file artifact was observed on removable media
- Archive activity occurred in the same experiment

### H4 — PARTIALLY_SUPPORTED (MEDIUM)
- Rename or deletion activity is consistent with possible concealment
- Artifact presence alone does not establish malicious intent

## Observation, Inference, Conclusion

**Observation:** Normalized artifact records describe device and file activity.

**Inference:** Temporal relationships are consistent with the generated correlation events.

**Conclusion:** The combination of device, filesystem, user-session, and application artifacts strongly supports the simulated data-transfer hypothesis. No single artifact independently esta blishes the complete transfer sequence.

## Experimental Score

Score: **13**. This is an experimental project model and is NOT an established forensic standar d.

## Limitations

- Simulated laboratory environment

- Artifact availability depends on Windows configuration

- Timestamp sources have different semantics and reliability

- USB presence alone does not establish transfer

- Correlation does not establish intent

- The experimental score is not a forensic standard

## Appendix — Disclaimer

This system is an educational digital-forensics research platform operating on a controlled lab oratory dataset. Findings represent an analytical interpretation of the supplied artifacts and should not be treated as a substitute for a validated forensic methodology in a real investigat ion.